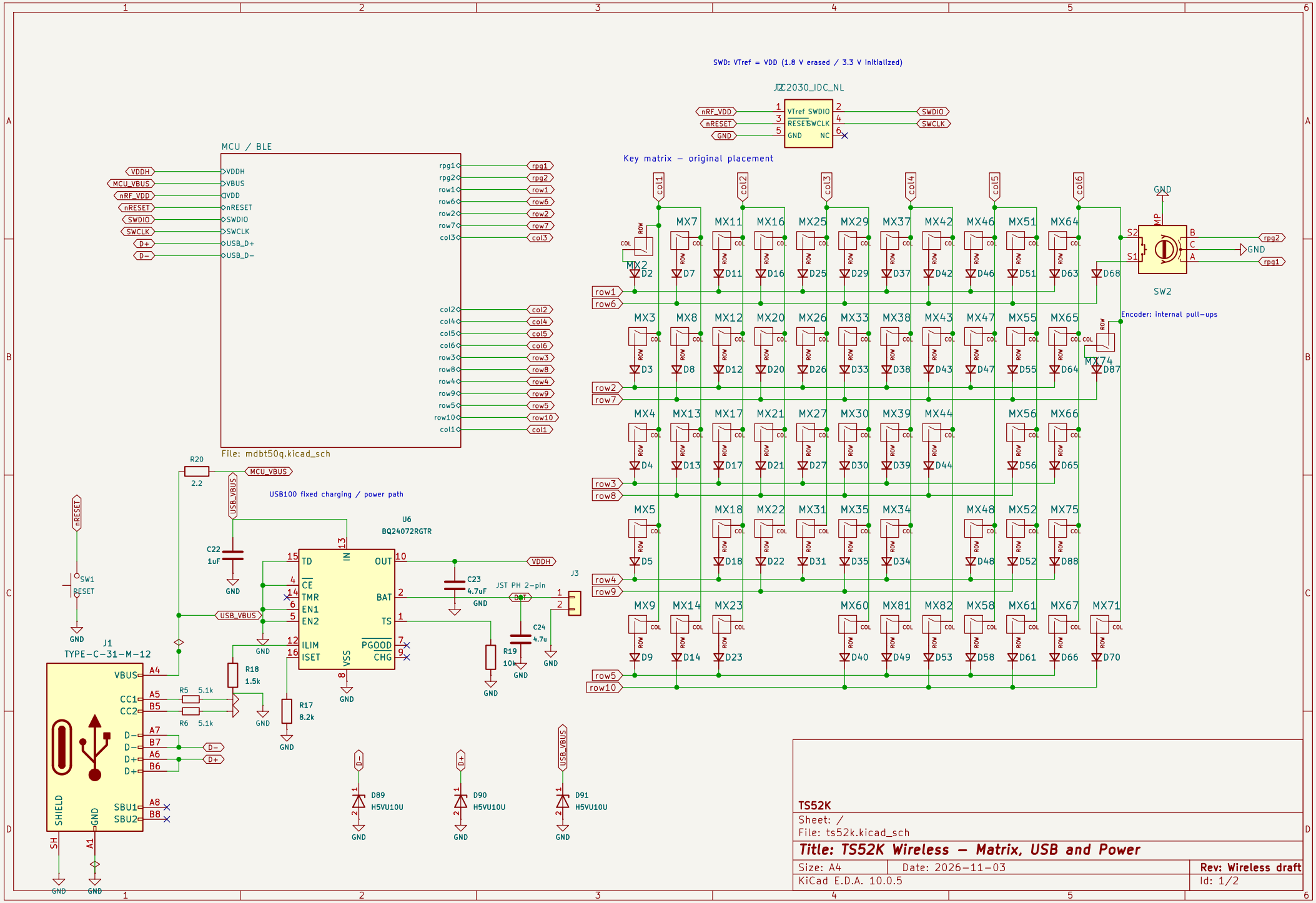
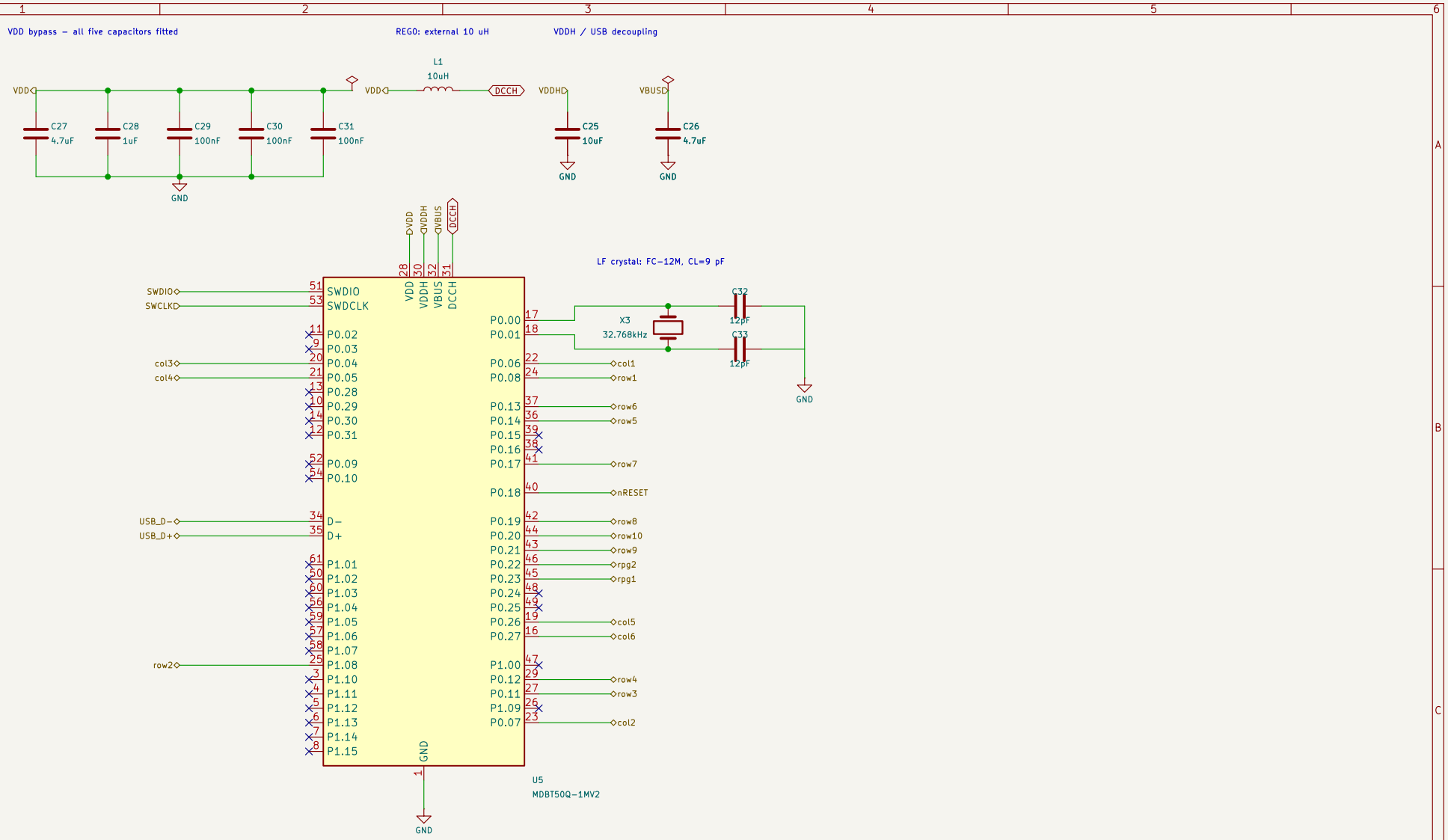

SWD: VTref = VDD (1.8 V erased / 3.3 V initialized)
JTC2030_IDC_NL
nRF_VDD
nRESET
GND
VTref SWDIO
RESET SWCLK
GND NC
SWDIO
SWCLK
MCU / BLE
VDDH
MCU_VBUS
nRF_VDD
nRESET
SWDIO
SWCLK
D+
D−
VDDH
VBUS
VDD
nRESET
SWDIO
SWCLK
USB_D+
USB_D−
rpg1
rpg2
row1
row6
row2
row7
col3
col2
col4
col5
col6
row3
row8
row4
row9
row5
row10
col1
File: mdbt50q.kicad_sch
Key matrix — original placement
col1
col2
col3
col4
col5
col6
MX7
MX11
MX16
MX25
MX29
MX37
MX42
MX46
MX51
MX64
MX2
MX3
MX8
MX12
MX20
MX26
MX33
MX38
MX43
MX47
MX55
MX65
MX74
MX4
MX13
MX17
MX21
MX27
MX30
MX39
MX44
MX56
MX66
MX5
MX18
MX22
MX31
MX35
MX34
MX48
MX52
MX75
MX9
MX14
MX23
MX60
MX81
MX82
MX58
MX61
MX67
MX71
D2
D7
D11
D16
D25
D29
D37
D42
D46
D51
D63
D68
D3
D8
D12
D20
D26
D33
D38
D43
D47
D55
D64
D87
D4
D13
D17
D21
D27
D30
D39
D44
D56
D65
D5
D18
D22
D31
D35
D34
D48
D52
D88
D9
D14
D23
D40
D49
D53
D58
D61
D66
D70
row1
row6
row2
row7
row3
row8
row4
row9
row5
row10
GND
MP
S2
S1
B
C
A
rpg2
GND
rpg1
SW2
Encoder: internal pull−ups
USB100 fixed charging / power path
U6
BQ24072RGTR
R20
2.2
MCU_VBUS
USB_VBUS
C22
1uF
GND
TD
CE
TMR
EN1
EN2
ILIM
ISET
IN
VSS
OUT
BAT
TS
PGOOD
CHG
VDDH
C23
4.7uF
JST PH 2−pin
J3
BAT
C24
4.7u
R19
10k
R18
1.5k
R17
8.2k
SW1
RESET
nRESET
J1
TYPE−C−31−M−12
VBUS
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
GND
SH
A1
R5
5.1k
R6
5.1k
A4
A5
B5
A7
B7
A6
B6
A8
B8
D89
H5VU10U
D90
H5VU10U
D91
H5VU10U
TS52K
Sheet: /
File: ts52k.kicad_sch
Title: TS52K Wireless — Matrix, USB and Power
Size: A4
Date: 2026−11−03
KiCad E.D.A. 10.0.5
Rev: Wireless draft
Id: 1/2


VDD bypass — all five capacitors fitted
REG0: external 10 uH
VDDH / USB decoupling
L1
10uH
VDD
DCCH
VDDH
VBUSD
C27
4.7uF
C28
1uF
C29
100nF
C30
100nF
C31
100nF
C25
10uF
C26
4.7uF
GND
LF crystal: FC-12M, CL=9 pF
X3
32.768kHz
C32
12pF
C33
12pF
SWDIO
SWCLKD
col3
col4
USB_D-
USB_D+
row2
col1
row1
row6
row5
row7
nRESET
row8
row10
row9
rpg2
rpg1
col5
col6
row4
row3
col2
U5
MDBT50Q-1MV2
P0.15 reserved for bootloader LED; leave unconnected.
32 MHz crystal and REG1 inductor are inside the module.
TS52K
Sheet: /MCU / BLE/
File: mdbt50q.kicad_sch
Title: TS52K Wireless — MDBT50Q and Clock
Size: A4
Date: 2026-11-03
Rev: Wireless draft
KiCad E.D.A. 10.0.5
Id: 2/2